

ABSTRACT

College Placement Analysis Using Python is a data analysis project developed to study and understand the placement performance of students using Python. The main objective of this project is to analyze placement data and identify the factors that influence student placements, such as CGPA, branch, internships, gender, and salary.

In this project, the placement dataset is imported from a CSV file and analyzed using Python libraries like **Pandas**, **NumPy**, and **Matplotlib**. Pandas is used for loading, cleaning, and organizing the data, NumPy is used for numerical calculations, and Matplotlib is used to create visualizations such as bar charts, pie charts, and histograms.

The project performs various analyses, including placement status distribution, branch-wise student count, gender-wise analysis, CGPA distribution, and salary analysis. It also checks for missing values, generates statistical summaries, and presents the results through graphical representations, making the data easy to understand.

The insights obtained from this project help identify important trends, such as the relationship between CGPA and placement, the impact of internships on placement opportunities, and salary distribution among placed students. These findings can assist students, faculty, and placement officers in understanding placement patterns and making informed decisions.

Overall, this project demonstrates how Python can be effectively used for real-world data analysis and visualization, transforming raw placement data into meaningful information that supports better academic and career planning.